CSE5026: Foundation and Frontiers of Cognitive Science

# Auditory and Speech Systems

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# Overview

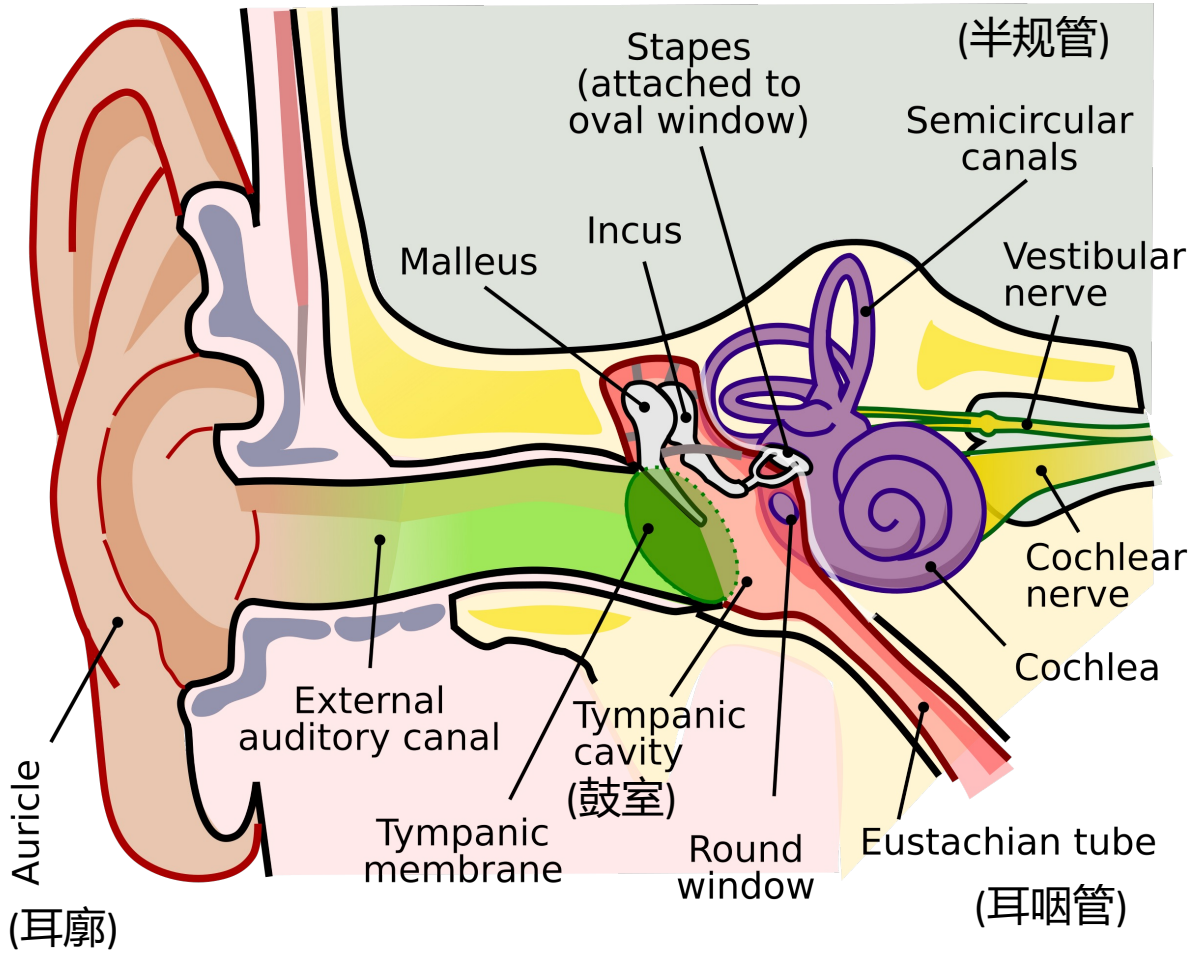
- Auditory Sensory System (听觉系统)
- Auditory Pathways (听觉神经通路)
- Speech Perception (言语感知)
- Speech Production (言语/语音生成)

# Auditory Perception (听觉感知)

- Sound waves excite sensory cells (感受细胞), triggering impulses in the auditory nerve (听神经). Auditory centers (听觉中枢) process these signals to produce hearing.
- Hearing involves mechanical motion, electrical changes, chemical transmission, neural impulses (神经冲动), and central processing.
  - 1. Sound collection by the outer ear and movement of the basilar membrane (基底膜) are mechanical.
  - 2. Hair cells (毛细胞) convert motion into electrical signals, releasing chemical transmitters and triggering neural impulses.
  - 3. The central nervous system (中枢神经系统) then processes the incoming information.

# The Auditory Process (听觉形成过程)

- Sound → outer ear (外耳) → middle ear (中耳) → cochlear receptors (耳蜗感受器) → auditory nerve (听神经)
- Auditory pathways (听觉传导通路) → auditory cortex (听觉皮层), where auditory perception develops
- External auditory canal (外耳道): 2.5–3.5 cm
- Tympanic membrane (鼓膜): 8–10 $mm^2$
- Cochlea (耳蜗)
- Vestibulocochlear nerve (前庭蜗神经): vestibular nerve (前庭神经) and cochlear nerve (耳蜗神经)




Source: https://en.wikipedia.org/wiki/Auditory_system

# The Auditory Process (听觉形成过程)



Malleus (锤骨)
Incus (砧骨)
Stapes (镫骨)

Cochlea (耳蜗): filled with fluid.

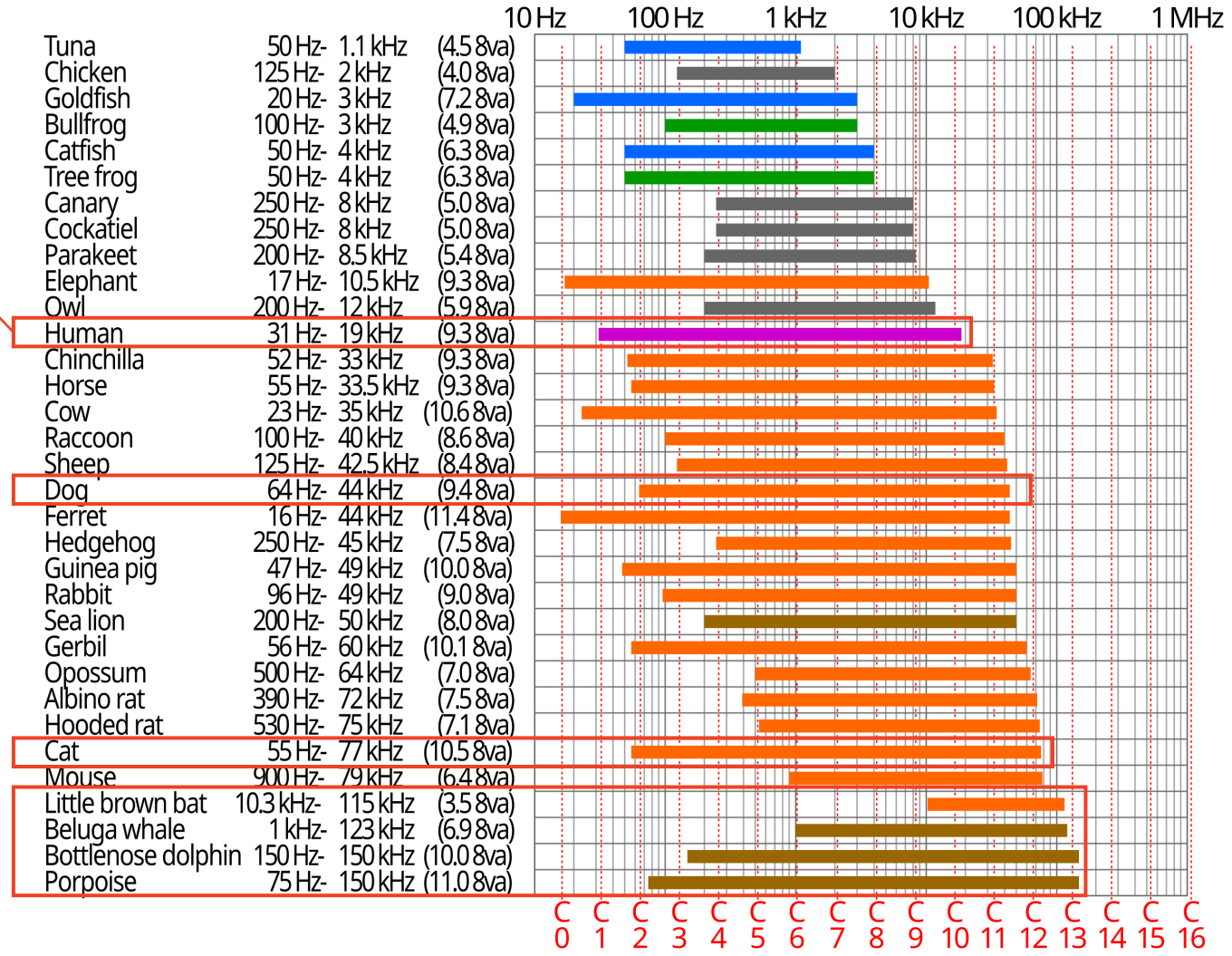
Sound vibrations create ripples in the fluid.

Stereocilia (静纤毛)
Hair cell (毛细胞)

Source: https://en.wikipedia.org/wiki/Hearing_range

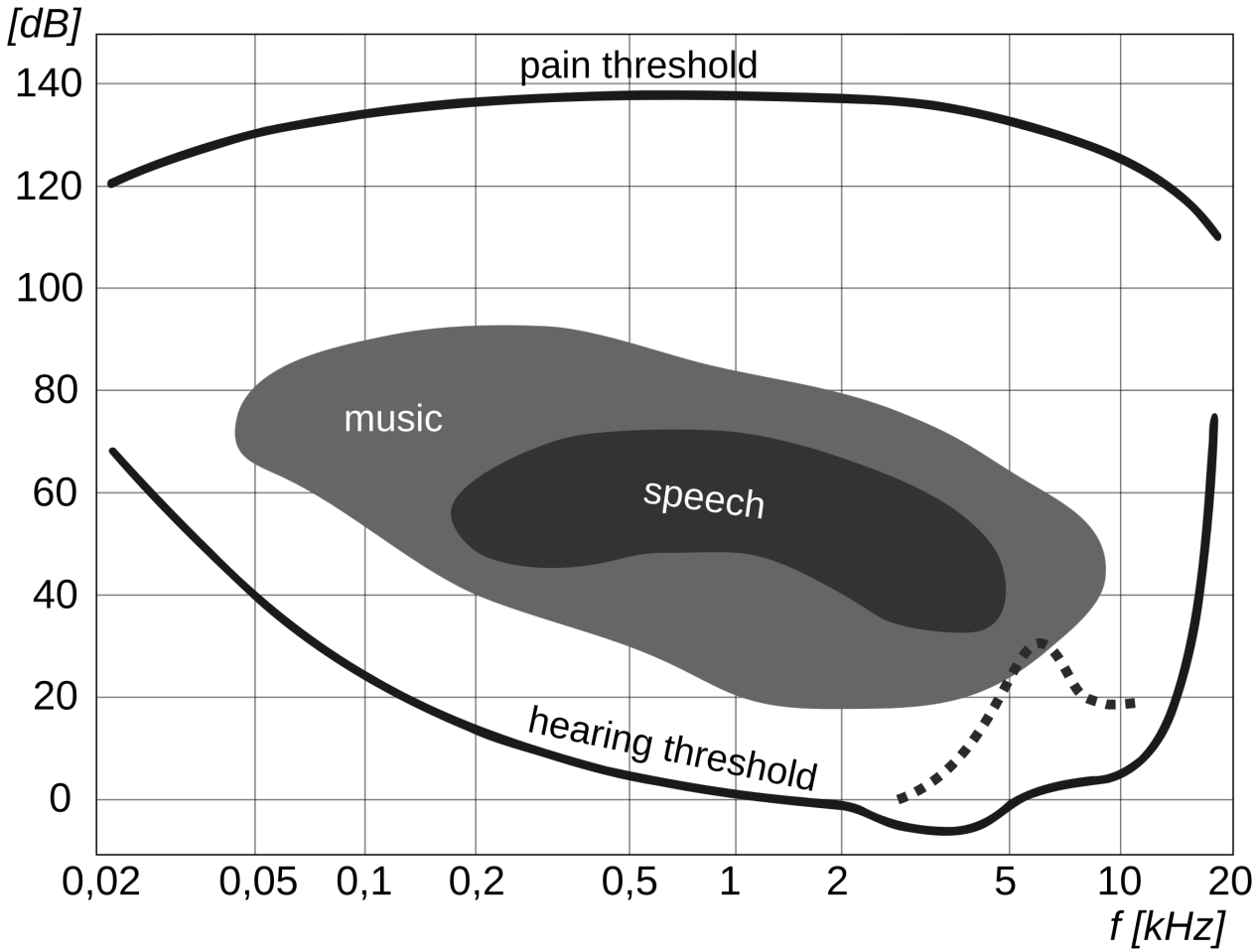
# Hearing Frequency Range (听觉频率范围)

- Humans: 20–20,000 Hz
- Dogs/cats: similar lower limits, higher upper limits (about 65,000 Hz)
- Bats: upper limit about 200 kHz
- Beluga whales: 1–123 kHz
- Bottlenose dolphins: 150 Hz–150 kHz



Source: https://en.wikipedia.org/wiki/Hearing_range

# Intensity–Frequency Range (强度–频率范围)




Measures of sound intensity

*Objective* (客观) measures:

- **Intensity** (声强)
- **S**ound **P**ressure **L**evel (声压级)

*Subjective* (主观) measure:

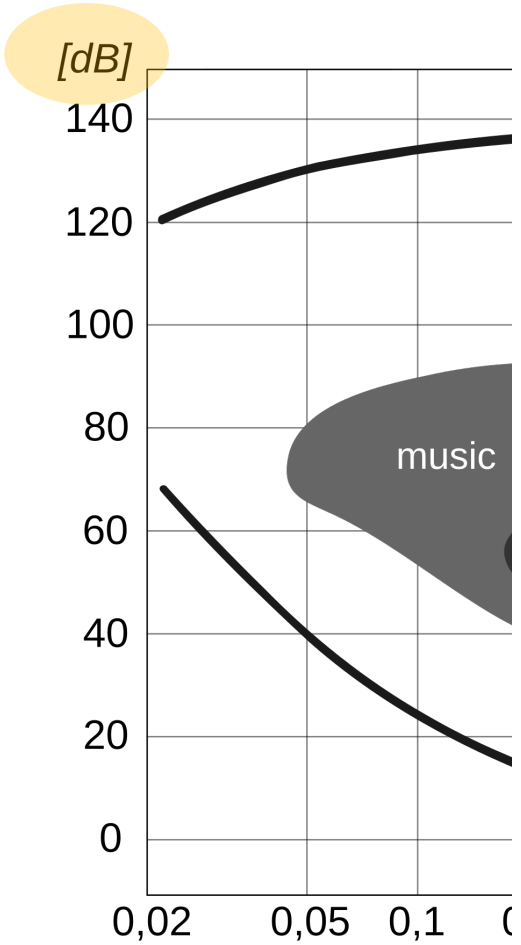
- **Loudness** (响度)

# Sound Pressure Level (声压级)

- SPL expresses the effective sound pressure on a logarithmic scale relative to a reference.

$$L_p = \ln\left(\frac{p}{p_0}\right) \text{Np} = 2\log_{10}\left(\frac{p}{p_0}\right) \text{B} = 20\log_{10}\left(\frac{p}{p_0}\right) \text{dB}$$

- The standard reference sound pressure in air is $p_0 = 20\mu$ Pa.
  - It approximates the human hearing threshold (听觉阈限), roughly the sound of a mosquito flying 3 m away.




Source: https://en.wikipedia.org/wiki/Sound_pressure

Source: https://en.wikipedia.org/wiki/Sound_pressure

# Examples of Sound Pressure Levels

| Sound source | Distance | Pa | dB |
|---|---|---|---|
| Jet engine | 1 m | 632 | 150 |
| Loudest human voice | 1 inch | 110 | 135 |
| Threshold of pain (疼痛阈限) | At the ear | 20–100 | 120–134 |
| Trumpet | 0.5 m | 63.2 | 130 |
| Baby crying | – | – | 80–130 |
| Normal conversation | 1 m | 2×10−3–0.02 | 40–60 |
| Very quiet room | | 2.00×10−4...6.32×10−4 | 20–30 |

# Review: Weber's Law (韦伯定律)

- Absolute threshold (绝对阈限): the minimum stimulus intensity needed to detect a single stimulus.
- Difference threshold (差别阈限): the smallest detectable difference between stimuli.
- Weber's law (韦伯定律): within a certain range, the difference threshold is approximately a constant fraction of stimulus intensity.

$$\frac{\Delta I}{I} = k$$

$\Delta I$: difference threshold; $I$: initial intensity

$k$ is the constant Weber fraction (韦伯分数)

# Weber's Law (韦伯定律)

- Typical Weber fractions $k$
  Initial stimulus intensities appear in parentheses.

$$\frac{\Delta I}{I} = k$$

$\Delta I$: difference threshold; $I$: initial intensity

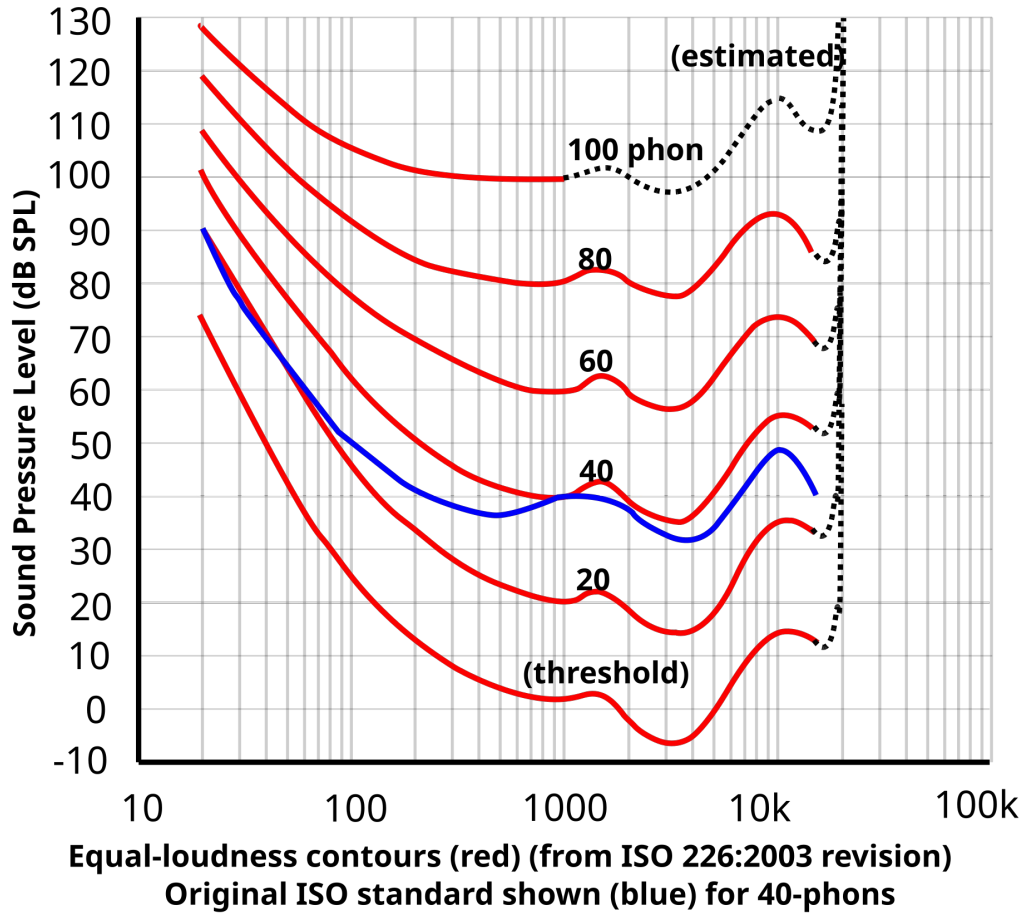
$k$ is the constant Weber fraction (韦伯分数)

| Stimulus (initial intensity) | Weber fraction |
|---|---|
| Pitch (2,000 Hz) | 1/333 |
| Deep pressure (400 g) | 1/77 |
| Visual brightness (1,000 photons) | 1/62 |
| Lifted weight (300 g) | 1/53 |
| Loudness (100 phon, 1,000 Hz) | 1/11 |
| Rubber odor (200 odor units) | 1/10 |
| Skin pressure (5 g/mm²) | 1/7 |
| Salt taste (3 gram-molecules/kg) | 1/5 |

# Loudness (主观响度)

- Loudness (响度) is the subjective perception of sound pressure. An audiometer can measure it using loudness balancing.
- Sensitivity to loudness varies with frequency. Equal-loudness contours (等响度曲线) show this relationship.






Equal-loudness contours (red) (from ISO 226:2003 revision)
Original ISO standard shown (blue) for 40-phons

Source: https://en.wikipedia.org/wiki/Loudness

# Loudness Level (响度级)

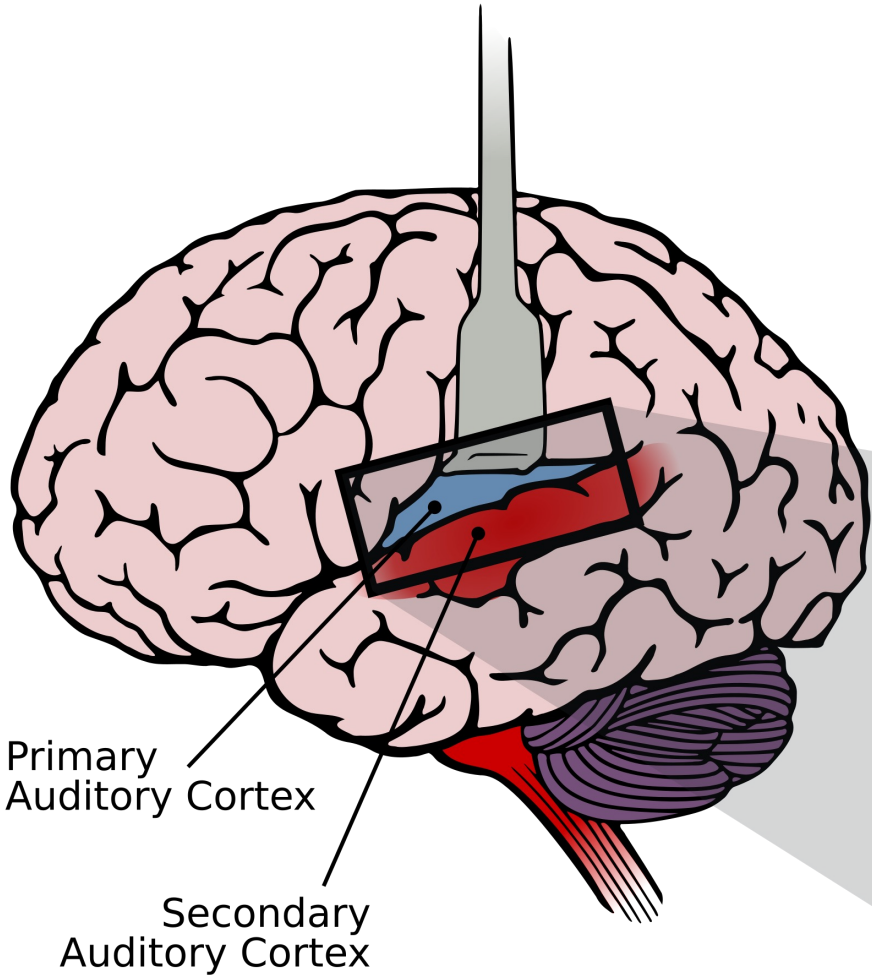
**Phon** is the unit of **loudness level** (响度级).

A sound has a loudness level of **x phon** if, under specified listening conditions, it sounds as loud as a **1,000 Hz reference pure tone** (参考纯音) at **x dB SPL**.

The labels on the **equal-loudness contours** (等响度曲线) on the previous slide use this same phon scale.

# Auditory Cortex (听觉皮层)

- The auditory cortex (听觉皮层) lies in the **temporal lobe (颞叶).**
- It includes primary auditory cortex (初级听觉皮层, A1), secondary auditory cortex (次级听觉皮层, A2), and association areas (联合区).
- It analyzes the timing and frequency of signals from the cochlea (耳蜗).
- Neurons (神经元) are organized by their preferred frequencies: low at one end and high at the other.
- Functional specialization: A1 encodes complex, abstract features of auditory stimuli, beyond raw features such as individual frequencies.

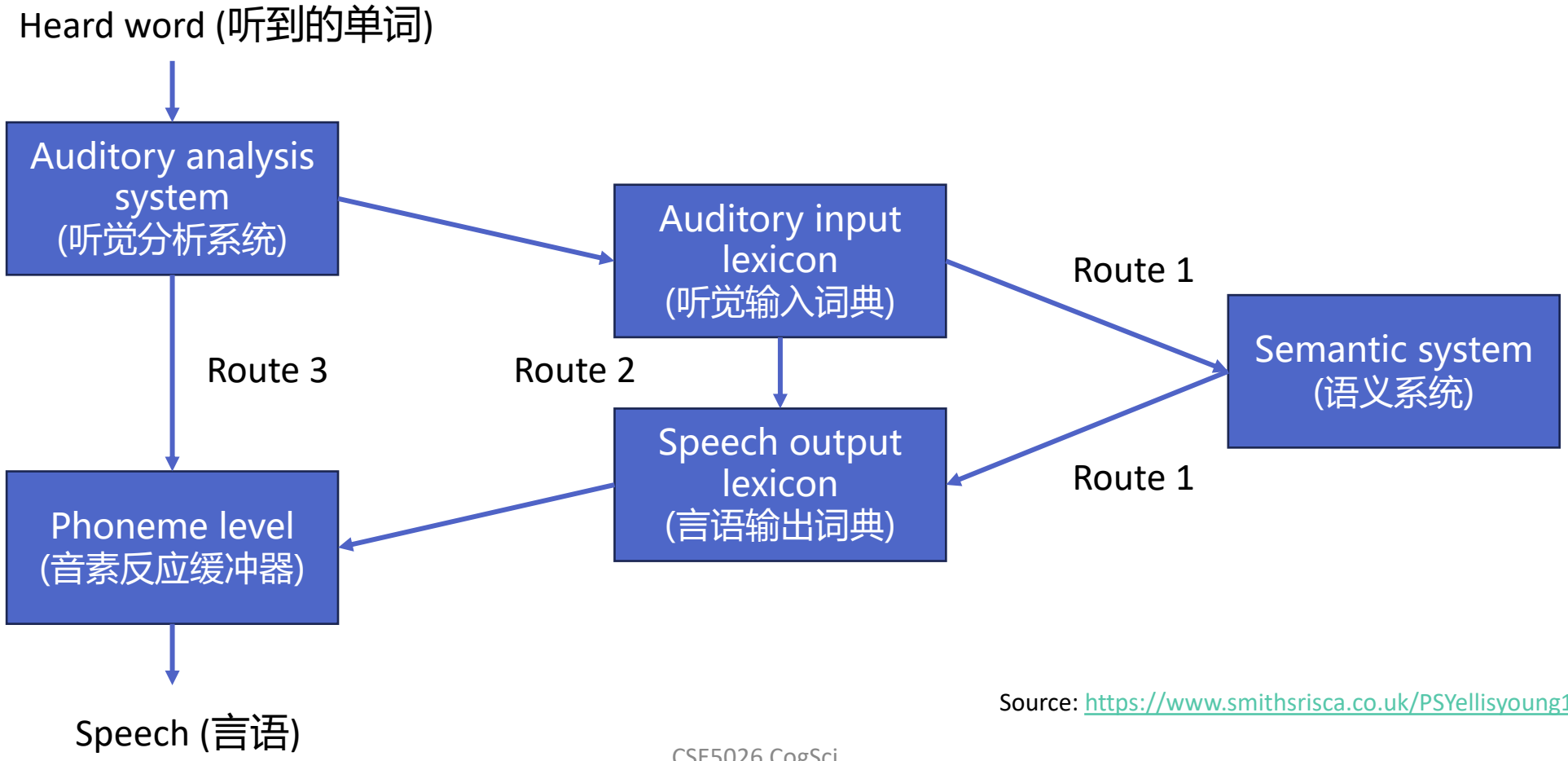


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- **Speech Perception (言语感知)**
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# Ellis & Young Model (口语通路模型)

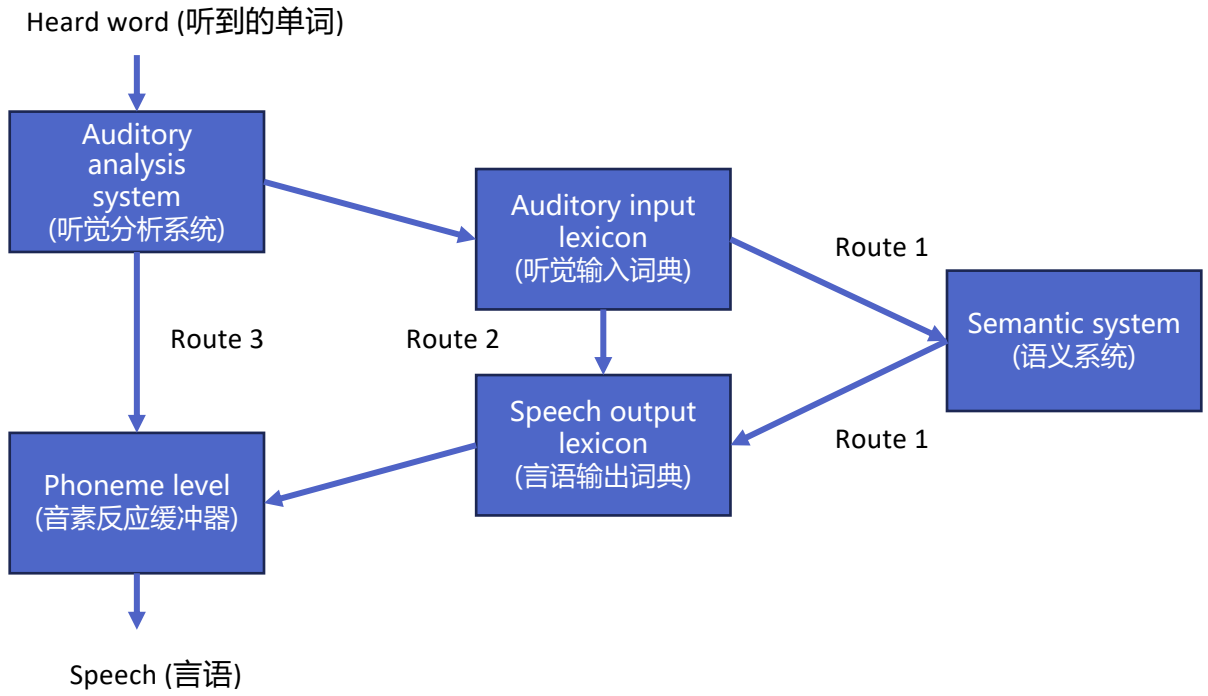
- Ellis & Young (1988): Speech perception and production




Source: https://www.smithsrisca.co.uk/PSYellisyoung1988.html

# Sound Analysis and Word Recognition

- Auditory analysis system (听觉分析系统): extracts phonemes and acoustic features.
  - Phoneme (音位/音素): the smallest unit distinguishing word meanings.
  - Auditory input lexicon (听觉输入词典): recognizes familiar spoken words.
    - Meaning comes next, in the semantic system (语义系统).

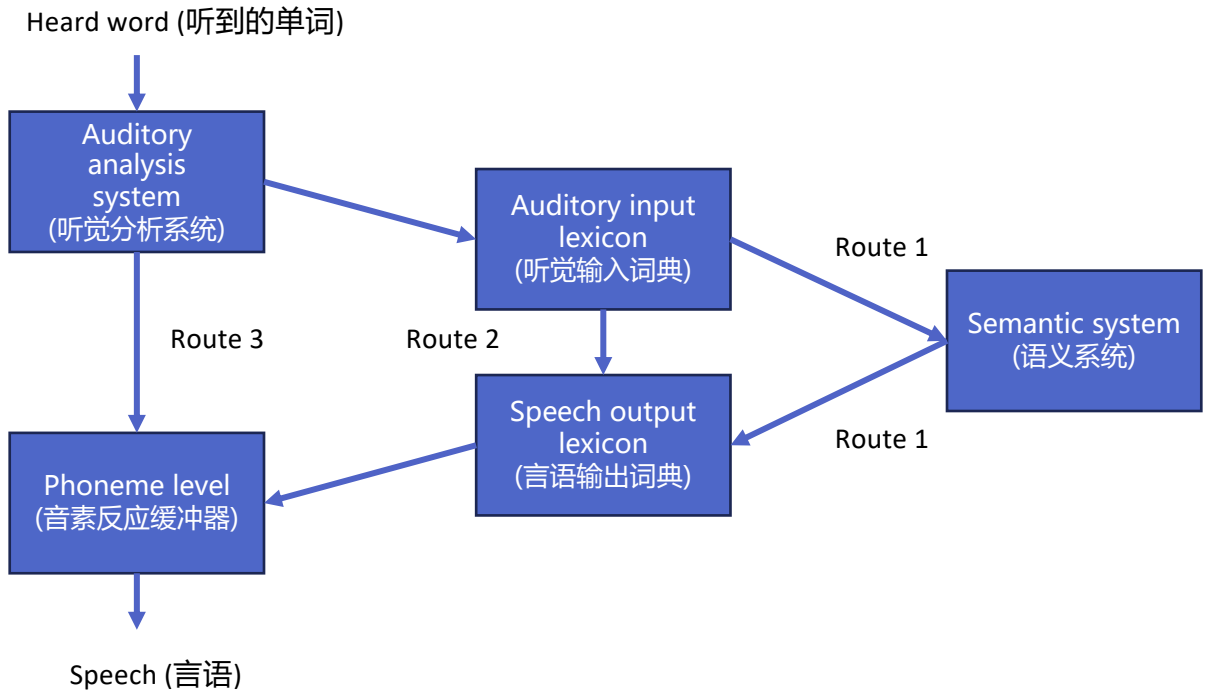



| | | |
|---|---|---|
| Unaspirated stop | /p/ | (爸, bà), (波, bō) |
| Aspirated stop | /pʰ/ | (怕, pà), (坡, pō) |

Source: Mandarin phoneme examples

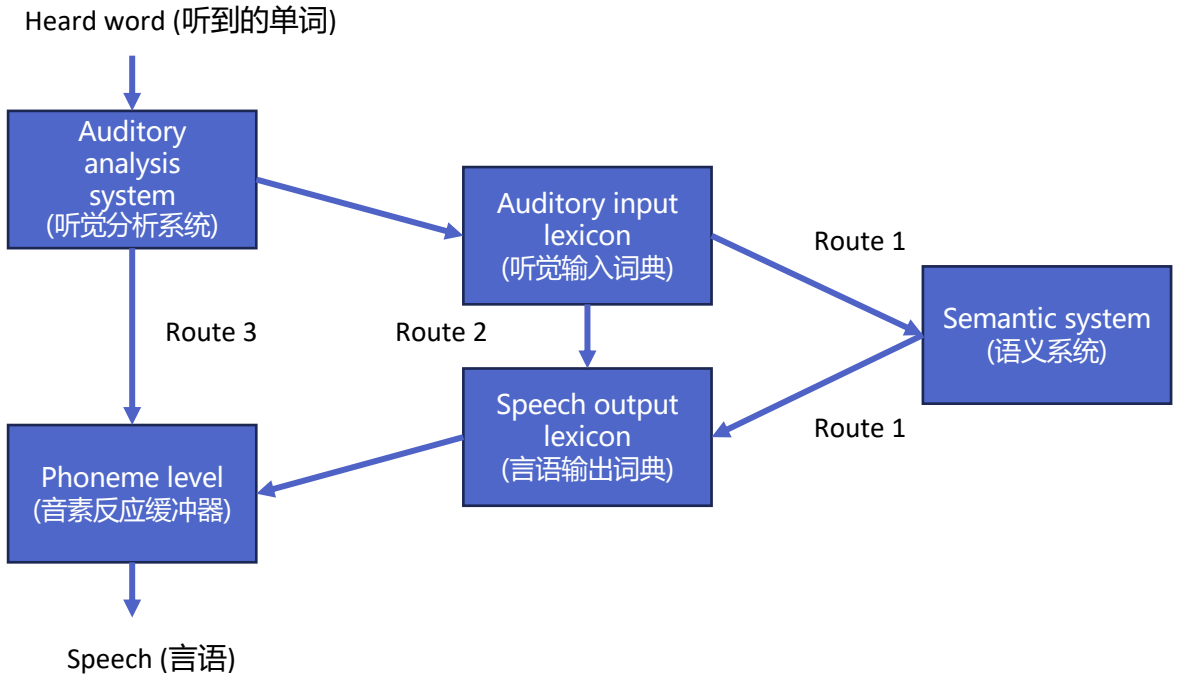
# Meaning and Speech Output

- Semantic system (语义系统): stores word meanings.
  - An abstract functional module. Its implementation could be distributed, as in semantic embeddings.
  - Speech output lexicon (言语输出词典): supplies stored spoken word forms.
  - Phoneme level (音素反应缓冲器): holds the speech sounds needed for output.



# Three Routes for Spoken Repetition

- Route 1 (semantic): recognizes a word form, accesses meaning, then retrieves speech output.
- Route 2 (lexical): repeats familiar words without accessing meaning.
- Route 3 (nonlexical): repeats unfamiliar words or nonwords via a direct link to the phoneme level.
- Word meaning deafness (词义性耳聋): impaired understanding of heard words with preserved repetition.



# Overview

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- **Speech Production (言语/语音生成)**

# Articulatory Phonetics (发音语音学)

- Articulatory phonetics studies how organs in the mouth, throat, and nose shape airflow from the lungs to produce speech sounds.
- Vocal organs (发声器官)
  - Vocal folds / cords (声带)
  - Closely spaced vocal folds vibrate as air passes through. Widely separated folds do not vibrate.
  - Voiced sounds (浊音) involve vocal-fold vibration, e.g., [b], [d], [g].
  - Unvoiced sounds (清音) lack vocal-fold vibration, e.g., [p], [t], [k].

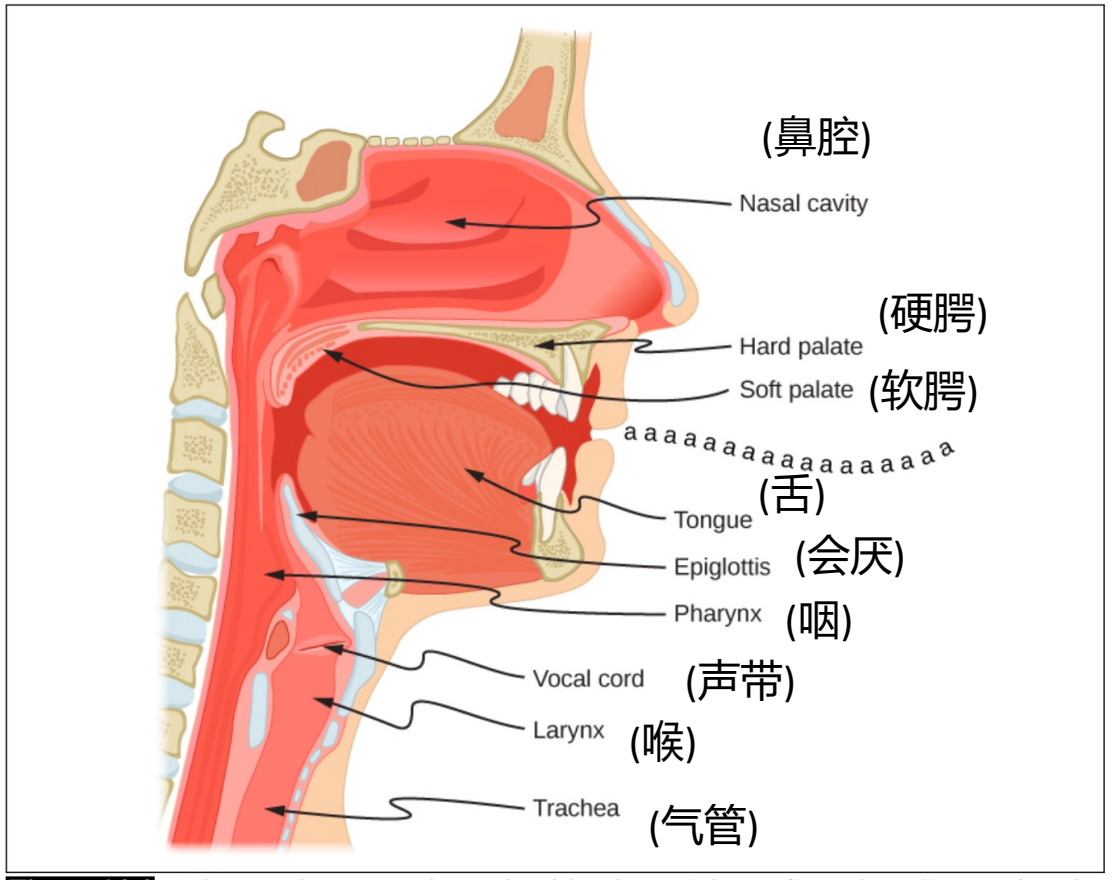
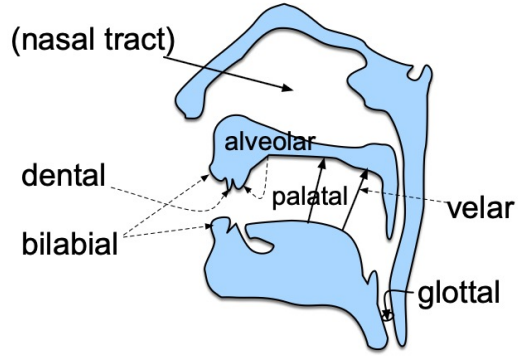



**Figure 14.4** The vocal organs, shown in side view. (Figure from OpenStax University Physics, CC BY 4.0)

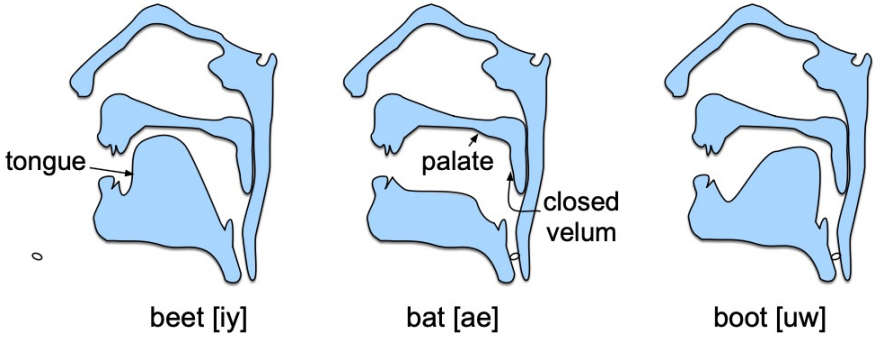
Source: https://web.stanford.edu/~jurafsky/slp3/14.pdf

# Consonants and Vowels (辅音与元音)

- Speech sounds fall into two broad classes: consonants and vowels.
- Consonants (辅音) restrict or block airflow and may be voiced or unvoiced.
- Vowels (元音) involve less obstruction. They are usually voiced, louder, and longer than consonants.
- Sounds can be classified further by place of articulation (发音部位).




Main places of articulation for English consonants




Tongue positions for different vowels

# Syllables (音节)

- Consonants and vowels combine to form syllables (音节).
- A syllable consists of a vowel-like sound (or sonorant) with its most closely associated consonants.
  - Example: “dog” has one syllable, [d aa g]. “Catnip” has two, [k ae t] and [n ih p].
- Nucleus (核音): the central vowel.
- Onset (首音): initial consonant(s).
- Coda (尾音): final consonant(s).
- Rime (韵) = nucleus + coda

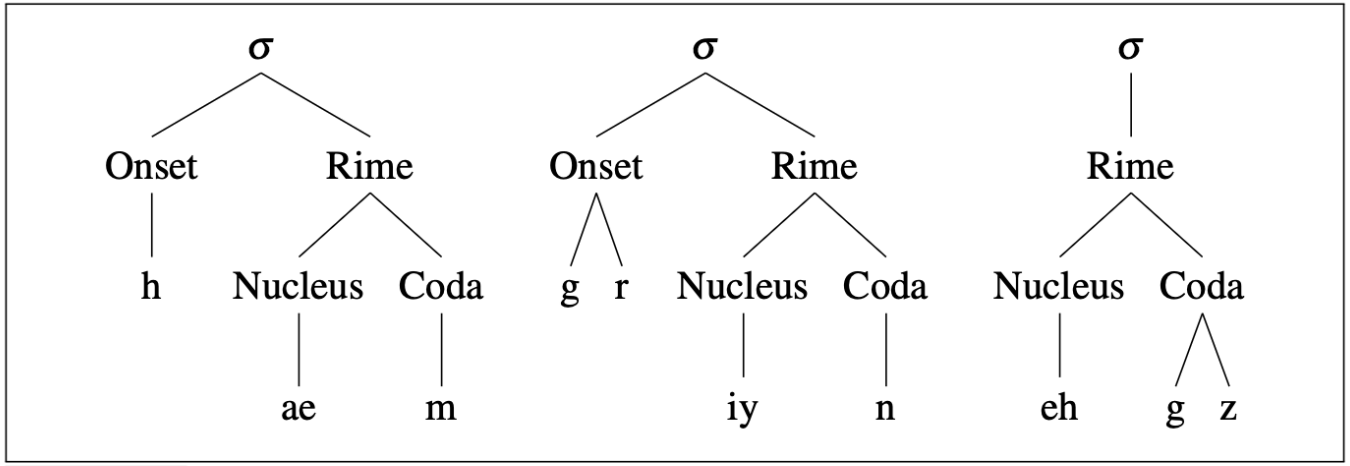



**Figure 14.8** Syllable structure of *ham*, *green*, *eggs*. σ=syllable.

# Prosody (韵律学)

- Prosody (韵律学) concerns intonation and rhythm, especially how **fundamental frequency (基频, F0)**, energy, and duration convey pragmatic, emotional, or conversational meaning.
- Energy relates to loudness (响度). F0 contributes to perceived pitch (音高).
- Prosody can mark discourse structure, including the distinction between statements and questions.
- Prosodic prominence involves pitch accent (音高重音), lexical stress (词汇重音), and vowel reduction to schwa (弱化元音/央元音).

# Prosodic Prominence (韵律突显)

- Pitch accent (音高重音): a prominent word or syllable carries an accent in pitch.
  - Example: accented words are underlined.

    (14.1) I'm a little surprised to hear it characterized as happy.

- Lexical stress (词汇重音): pitch accents occur on syllables that can bear stress.
  - Underlined words carry accent. Bold syllables carry lexical stress.

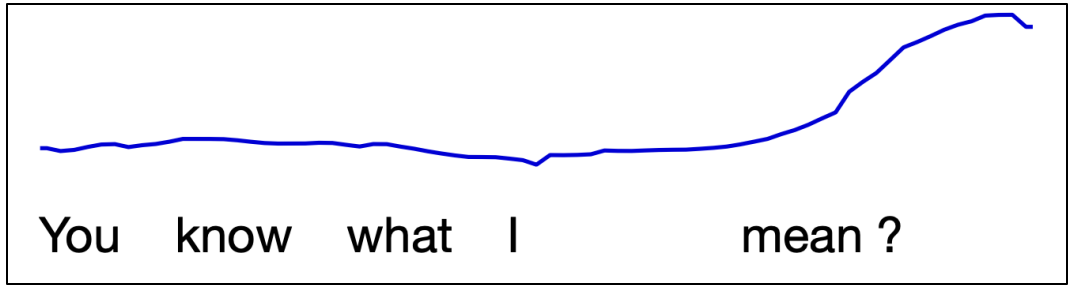
    (14.2) I'm a little sur**prised** to hear it **char**acterized as **hap**py.

  - Lexical stress is specified in the lexicon.

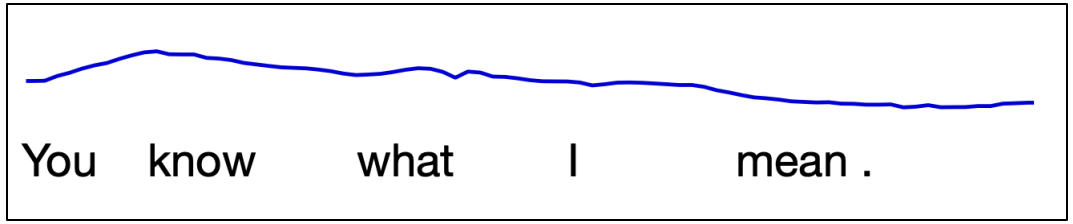
- Unstressed vowels may reduce to schwa (弱化元音/央元音).

# Intonation (语调)

- Utterances with the same accents may still differ in prosody because they have different intonation patterns.
- Intonation (语调) is the *rise* and *fall* of **fundamental frequency (基频, F0)** over time.
- Example: statements and yes–no questions




A final rise in F0 indicates
a yes–no question (question rise)




A final fall in F0 indicates
a statement (final fall)

# Basics of Speech Signal Processing

- Sound waves are characterized by frequency (频率) and amplitude (振幅).
- $y = A * \sin(2\pi f t)$: a sine wave with amplitude $A$ and frequency $f$

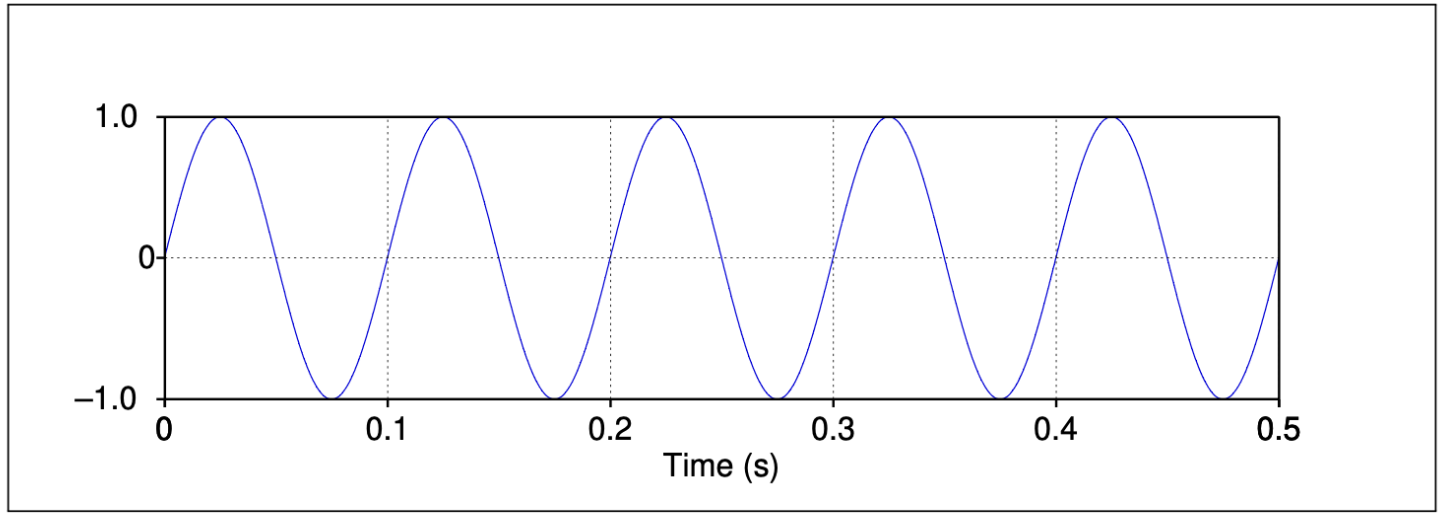



**Figure 14.10** A sine wave with a frequency of 10 Hz and an amplitude of 1.

$f = 10\text{Hz}$ => 10 cycles per sec

# Speech Waveforms

- A waveform plots air pressure as it changes over time.

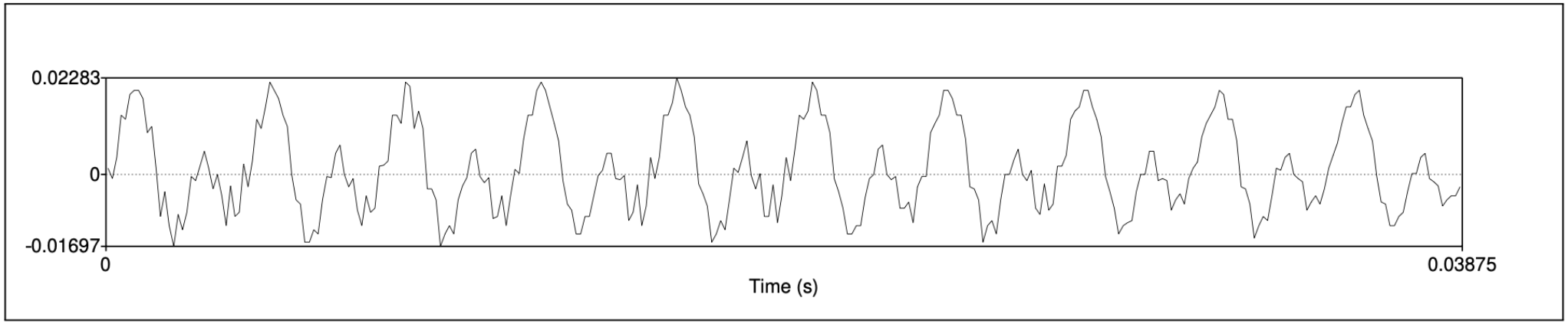



**Figure 14.11** A waveform of the vowel [iy] from an utterance shown later in Fig. 14.15 on page 15. The *y*-axis shows the level of air pressure above and below normal atmospheric pressure. The *x*-axis shows time.

Positive values indicate **compression**. Negative values indicate rarefaction.

# From Sound Waves to Digital Signals

- Analog-to-digital conversion has two main steps: sampling and quantization.
- Sampling rate (采样率), $f_s$: the number of samples per second. To avoid aliasing, signal frequencies must stay below the Nyquist frequency, $f_N = \frac{1}{2} f_s$.
- Most speech information lies at frequencies $\leq 10000$ Hz. Preserving that bandwidth requires a sampling rate of at least $f_s = 20000$ Hz.
- Traditional landline telephony: 8 kHz. A common audio sampling rate for online video: 48 kHz.

# From Sound Waves to Digital Signals

- Even at an 8 kHz sampling rate, one second of speech requires 8,000 amplitude measurements. Efficient storage matters.
- Quantization (量化) stores amplitude as integers: 8-bit (−128 to 127) or 16-bit (−32,768 to 32,767).
  - Traditional telephony: 8 kHz sampling and 8-bit quantization.
- $\mu$-law PCM (pulse code modulation, 脉冲编码调制):

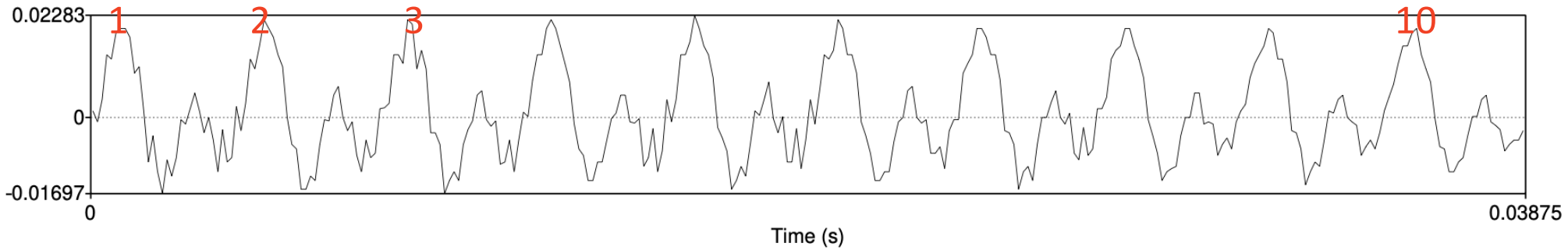
$$F(x) = \frac{\operatorname{sgn}(x)\log(1+\mu|x|)}{\log(1+\mu)} \quad -1 \le x \le 1$$

(μ=255 for 8 bits)

- Logarithmic compression reflects the greater sensitivity of human hearing to changes at low intensities.

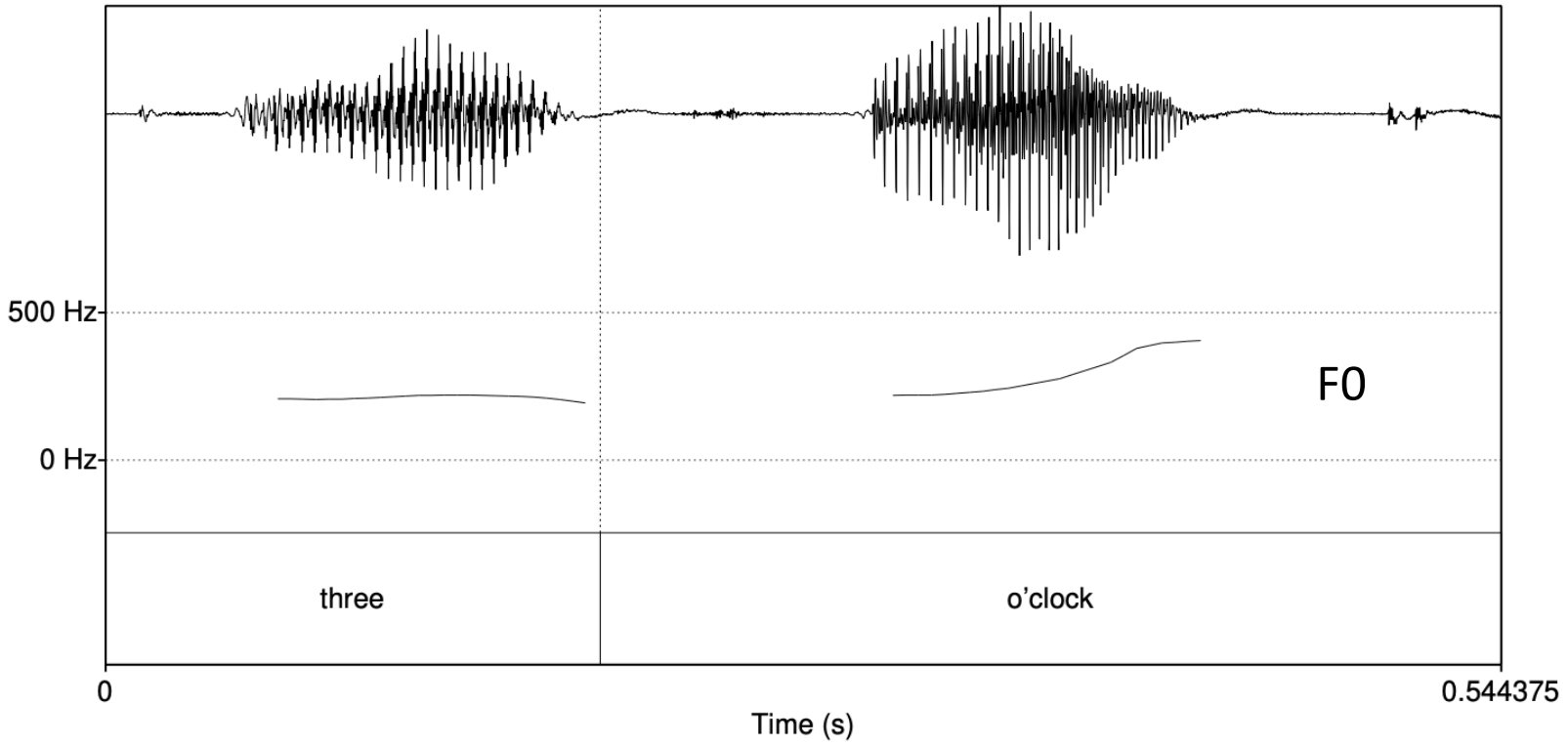
# Frequency and Amplitude of Speech

- A speech waveform is not sinusoidal, but voiced speech shows periodicity.




- Ten similar peaks occur in 38.75 ms: $T = \frac{10}{0.03875} \approx 258\ \mathrm{Hz}$
- Voicing comes from repeated opening and closing of the vocal folds (声带). Opening releases pressurized air from the lungs, creating a pressure pulse.
- Each major peak corresponds to one cycle of vocal-fold vibration. This repetition rate is the fundamental frequency (基频), F0.

# Frequency and Amplitude of Speech




A pitch track (音高轨迹) plots fundamental frequency (基频, F0) over time.

Here, F0 denotes a frequency value (0–500 Hz), rather than the amplitude of the fundamental component.

For “Three o’clock?”, F0 rises at the end of the question.

# Frequency and Amplitude of Speech

- Besides instantaneous amplitude, we often need a measure of amplitude over a time window.
- Signal power is the mean of the squared amplitudes:

$$\text{Power} = \frac{1}{N}\sum_{i=1}^{N} x_i^2$$

- Intensity in decibels (dB) expresses power relative to a reference corresponding to the human hearing threshold (听觉阈限).

$$\text{Intensity} = 10\log_{10}\frac{1}{NP_0}\sum_{i=1}^{N} x_i^2$$

$P_0 = 2\times10^{-5}$ Pa
(reference pressure)

# Pitch and Loudness (音高与响度)

- Pitch (音高) relates to frequency. Loudness (响度) relates to intensity.
- Pitch is the perceptual counterpart of fundamental frequency (基频, F0).
  - Higher F0 generally produces a higher perceived pitch.
  - The relationship is nonlinear because auditory sensitivity varies across frequencies.
  - Between about 100 and 1,000 Hz, pitch perception is most accurate and approximately linear in frequency.
  - Above 1,000 Hz, the relationship is approximately logarithmic.
- Mel scale model (梅尔标度):

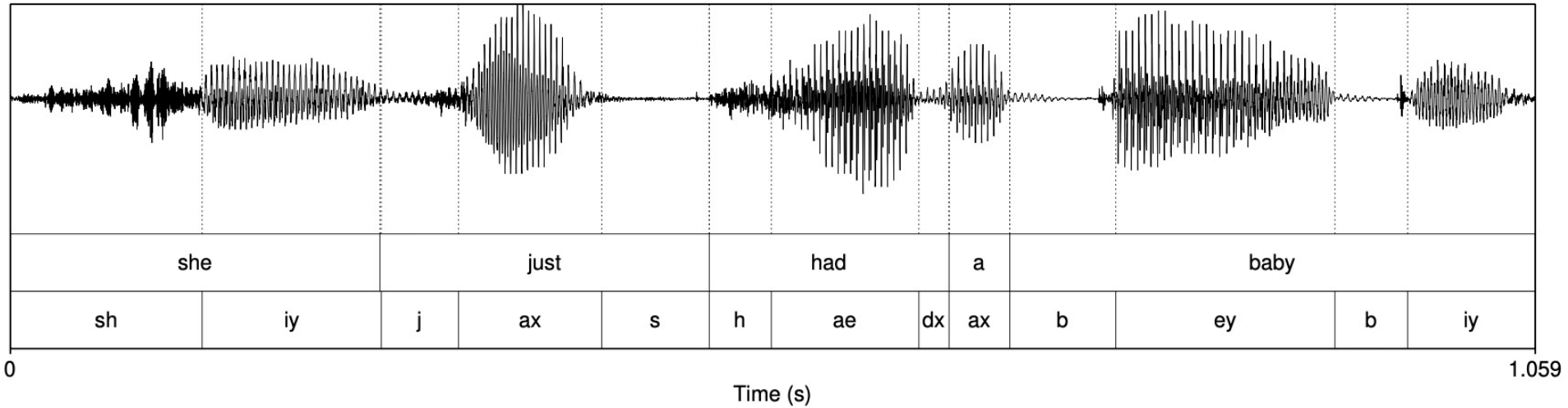
$$m = 1127 \ln(1 + \frac{f}{700})$$

Equal perceived pitch intervals correspond to equal differences in mel values.

Source: https://web.stanford.edu/~jurafsky/slp3/14.pdf

# Interpreting Speech Waveforms

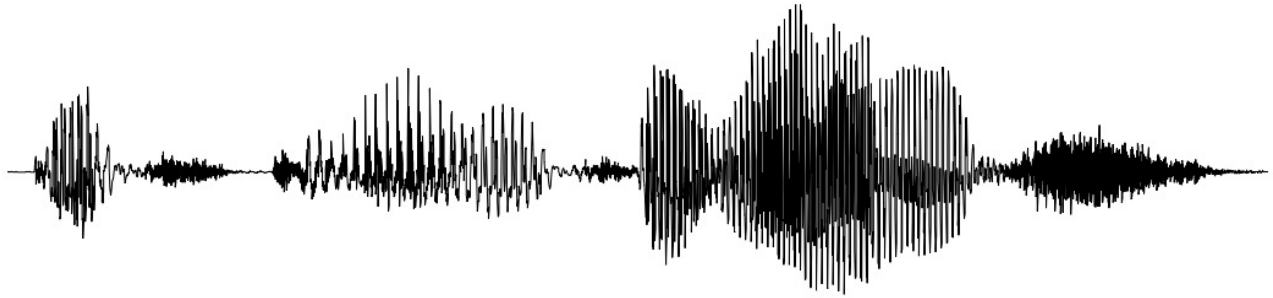
- Vowels (元音) are easier to identify because they are relatively long and loud.
  - The six vowels [iy], [ax], [ae], [ax], [ey], and [iy] show regular amplitude peaks, indicating voicing (浊音).
- Stop consonants (塞音) often show a near-silent closure followed by a small burst.
  - Two [b] sounds in “baby”



Source: https://web.stanford.edu/~jurafsky/slp3/15.pdf

# Automatic Speech Recognition (语音识别)

- ASR maps a speech waveform like this:



- to the corresponding words: “It’s time for lunch!”
- A ConvNet can extract acoustic features for an encoder–decoder model.
- The reverse task is text-to-speech (TTS, 语音合成).